Where Does a Language Model Keep What It Knows About Relevance? Attention Heads Decode Retrieval Relevance Better Than the Output Representation

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Abstract

Ask a language model whether a retrieved passage answers a query and its zero-shot verdict is the weakest of the four relevance signals we can extract from the same forward pass. We compare four *readers* that all consume the same frozen forward pass over the same prompt and the same answer token: the model's literal verdict, a trained linear probe on the final-layer hidden state (the output representation), a trained probe on the best residual-stream layer, and a probe on the top attention heads. Across eight instruction- and base-tuned models from three families (LLaMA, Mistral, Qwen; 3B–8B; 1500 queries each), reading the attention heads decodes relevance better than reading the output representation on *every* model. The advantage is +0.06 to +0.15 AUC, with query-clustered bootstrap confidence intervals that exclude zero after Benjamini–Hochberg correction across the full 8×4 family of tests ($q \leq 0.001$ throughout)—it is the only one of the four contrasts we test that survives correction on all eight models. The edge is not an artefact of dimensionality: matching the final layer's read-out procedure to the head probe's feature budget leaves it essentially unchanged. It is also not an artefact of training: fitting a probe on the output representation never separates from the model's own zero-shot verdict after correction (on seven of eight models the judge is in fact the stronger of the two). We are explicit about where the claim stops. The comparison is defined against one particular output-side reader, and re-reading that same layer with a different read-out (a per-chunk ensemble) closes or reverses the edge on two of eight models; and the residual-stream version of the comparison is significant on the three base models only. What we can say without qualification is this: for the task of judging retrieval relevance, the attention heads of a frozen LLM are the best place we found to read the answer, and the model's own output is the worst.

1 Introduction

A retrieval-augmented system hands a language model a passage and a question. The model must decide whether the passage is relevant. In our experiments it usually can—just not where you would look first.

Consider what the same forward pass offers. The model emits a verbal verdict, and reading the yes/no logit difference is free but weak. A logistic regression on the final-layer hidden state does better, because it is trained on labelled relevance. But the final layer is only one place in a deep stack, and it is the place furthest along a computation that must also serve every other task the model performs. We asked a narrow, measurable question: *holding the model, the prompt, the*

token position, and the training budget fixed, which internal representation is the best place to read relevance from?

The answer, across eight models, is the attention heads. This is a *relative* result and we keep it that way. We do not claim that a probe beats a trained cross-encoder reranker; an earlier phase of this project tested that and found it false, and we do not revisit it. We do not claim cross-domain generalisation. We claim something smaller and, we think, more useful: that the choice *where* to read is worth +0.06 to +0.15 AUC, reliably, across model families and training regimes.

Contributions.

- **A controlled four-way comparison** (§2) on one frozen forward pass, with layer and head selection done on a held-out validation split, and with all four readers reporting over identical rows.
- **A result that replicates across the panel** (§3): attention heads beat the output representation on 8/8 models, and they are the only one of the four contrasts that survives Benjamini–Hochberg correction everywhere.
- **Two controls that rule out the obvious explanations** (§4): a capacity-matched read-out control that leaves the edge intact, and a training control showing that a probe on the output representation adds nothing over the zero-shot verdict.
- **An explicit statement of where the claim stops** (§5), including the read-out sensitivity we found on two models and the fact that the residual-stream form of the comparison holds only on base models.

What we deliberately do not claim. This is a probing study, so every reader here is correlational. We initially expected to add a causal account of *why* the output is weak, and we ran that experiment; it did not support the claim we hoped for, and we report the null in the companion note rather than dress it up. The present paper therefore makes no mechanistic assertion. It is about measurement.

2 Setup

2.1 Task and data

We use MS MARCO v1.1, forming (query, passage, label) triples. Each query contributes its relevant passages (as marked in the dataset) and hard negatives drawn from the same query's non-selected retrieved passages, so "irrelevant" means *retrieved but not judged relevant* rather than *random*. Splits are made *by query*, 70/15/15, so no query appears in two splits. We sample 1500 queries, giving 7456 triples: 5218 train (1050 queries), 1121 validation (225), and 1117 test (225; 245 relevant rows).

Three disciplines keep the comparison honest. First, **layer and head selection use the validation split only**; the test split is touched once, and every per-layer or per-head maximum we report is a validation choice. Fields whose names carry `DO_NOT_REPORT` exist to make the test-selected alternative explicit and excluded. Second, all four readers are evaluated on **identical rows** with identical labels; we assert both label and query-id agreement against the activation cache rather than trusting a shared index. Third, every reader uses the **same regularisation strength** (C=1.0), so the four columns of Table 2 are mutually consistent and their differences reconcile with the reported edges.

Table 1: Feature geometry. The head probe uses fewer features than the residual probe on every model, so its advantage cannot come from dimensionality.

Model family	layers	heads	d_{head}	hidden	head probe (k=20)
LLaMA-3.2-3B	28	24	128	3072	2560
LLaMA-3.1-8B	32	32	128	4096	2560
Mistral-7B	32	32	128	4096	2560
Qwen2.5-7B	28	28	128	3584	2560

2.2 Prompt and activation extraction

Every reader sees the byte-identically identical prompt

```
Document: {passage}\nQuestion: {query}\nIs the document relevant to the question?
Answer:
```

and every activation is taken at the final *answer token*. We enforce this identity mechanically: a self-check script reconstructs the prompt from each of the four code paths that build it and fails if the strings differ.

For each decoder layer i we cache (a) the residual stream, i.e. `layers[i].output[0,-1,:]`, and (b) the per-head attention outputs, obtained by reshaping `self_attn.o_proj.output[0,-1,:]` to $[n_{\text{heads}}, d_{\text{head}}]$. All three model families in our panel have $d_{\text{head}} = 128$ (`hidden` $= n_{\text{heads}} \times 128$), so the per-head representation has the same geometry throughout. Activations are cached in `float16`.

2.3 Four readers

1. **LLM-judge** (*zero training*). The normalised P(yes) vs. P(no) at the answer token, summed over five surface variants of each word (`yes`/`Yes`/`YES`/`" yes"`/`" YES"`). No labels are used.

2. **Output-representation probe**. An ℓ_2-regularised logistic regression on the final-layer residual, fitted on train.

3. **Internal-residual probe**. The same probe on the validation-selected residual layer L^*.

4. **Attention-head probe** (*ours*). Per-head validation AUC ranks all $n_{\text{layers}} \times n_{\text{heads}}$ heads; we take the top k=20, standardise each head's activations, concatenate, and fit one logistic regression on train.

The head probe is deliberately simple. We are not proposing an architecture; we are asking which of four equally simple ways of reading the same forward pass recovers the most relevance. A head-space budget of $20 \times 128 = 2560$ dimensions is *smaller* than the residual stream (3072–4096 dimensions), so the head probe is not winning on feature count (Table 1).

2.4 Statistics

Because the split is by query, the resampling unit is the query. We report 95% confidence intervals and one-sided p-values from a *query-clustered paired bootstrap* (5000 resamples): the same resampled query set is used for both readers in a contrast, which is what makes the interval a statement about their difference rather than about two independent estimates. With 5000 resamples the smallest attainable p is 2×10^{-4}, and we write $p < 2 \times 10^{-4}$ rather than 0.

Table 2: Four readers on the same frozen forward pass (test AUC, 1500 queries per model; layer and head selection on validation; one regularisation strength C=1.0 for all readers). **Bold**: best reader in the row. The attention-head probe is best on all eight models. Figure 1 shows the same numbers with confidence intervals.

Model	L	judge	final	best-resid	attn (ours)
LLaMA-3.2-3B (base)	28	0.545	0.536	0.613 (L9)	**0.684**
LLaMA-3.1-8B (base)	32	0.574	0.562	0.621 (L8)	**0.677**
Mistral-7B-v0.3 (base)	32	0.579	0.583	0.606 (L16)	**0.684**
LLaMA-3.2-3B-Instruct	28	0.614	0.612	0.638 (L18)	**0.712**
LLaMA-3.1-8B-Instruct	32	0.630	0.596	0.609 (L12)	**0.725**
Mistral-7B-Instruct-v0.3	32	0.646	0.596	0.634 (L13)	**0.713**
Qwen2.5-7B (base)	28	0.652	0.592	0.655 (L22)	**0.680**
Qwen2.5-7B-Instruct	28	0.660	0.610	0.630 (L23)	**0.669**

We test four contrasts over eight models, so we report Benjamini–Hochberg q-values computed within each contrast family (8 tests). The headline claim of this paper is the one that survives that correction everywhere.

Like every result here, the statistics are derived by scripts from raw JSON artifacts, and a single collection script rebuilds the paper's number table with a provenance path recorded for every value. We mention this because our own earlier draft of this work quoted two different values for the same quantity in the same table; the remedy was mechanical rather than editorial.

3 Results

3.1 The attention heads are the best reader, on every model

Table 2 and Figure 1 give the full picture. Two things stand out.

First, **the zero-shot verdict is the weakest reader on all eight models**. It ranges from 0.545 (LLaMA-3.2-3B) to 0.660 (Qwen2.5-7B-Instruct) and is below the head probe everywhere. It is *not*, however, below the trained output-representation probe: the judge is the stronger of the two on seven of eight models and the weaker on one (Mistral-7B, 0.579 vs. 0.583). Training a probe on the final layer buys very little over asking the model—which is the point of the `final − judge` row in Table 3—and that is precisely why the interesting contrast is the *head* probe against the output representation, not the judge against anything. A model that can decode relevance at 0.67–0.73 from its attention heads is answering at 0.55–0.66 when asked directly.

Second, **the attention-head probe is the best reader on all eight models**. It beats the output-representation probe by +0.059 to +0.148 AUC, and it beats the best residual-stream layer on all eight as well, though that second margin is smaller.

3.2 The edge survives correction, and it is the only one that does

Figure 2 shows the head-probe-minus-output-representation edge per model, with intervals and corrected q-values. It lies between +0.059 and +0.148 and excludes zero after correction on all eight models.

Table 3 puts that in context by reporting all four contrasts we tested. Only one survives on every model. The residual-stream contrast (best-resid − final) survives on the three base models and loses significance on their instruct counterparts; the contrast against the judge (best-resid −

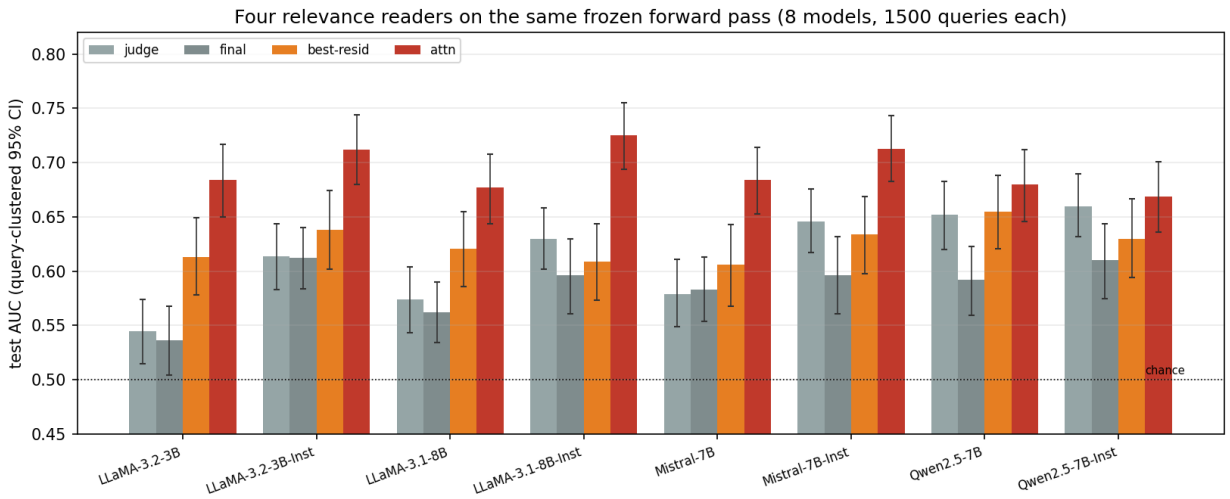


Figure 1: The four readers across eight models, with query-clustered 95% confidence intervals. The head probe (red) is highest in every group; the zero-shot judge (light grey) is lowest or near-lowest everywhere.

Table 3: All four contrasts, with Benjamini–Hochberg q within each family (8 tests). A cell is marked • when the 95% CI excludes zero after correction. The attention-head contrast is the only one that survives everywhere.

Contrast	survives (of 8)	range of effect	models where it survives
attn − final	**8**	$+0.059 \ldots + 0.148$	all eight
best-resid − final	3	$+0.059 \ldots + 0.078$	LLaMA-3.2-3B, LLaMA-3.1-8B, Qwen2.5-7B (base)
best-resid − judge	2 (1 after BH)	$+0.047 \ldots + 0.068$	LLaMA-3.2-3B, LLaMA-3.1-8B
final − judge	0	$-0.060 \ldots + 0.004$	—

judge) excludes zero on two models and survives correction on one; and the "pure training" contrast (final − judge) survives on none. The last of these is worth emphasising, because it is the control that matters most for the framing of this paper: *training a probe on the output representation does not beat simply asking the model*, on any model in our panel.

Figure 3 presents the same comparison two other ways: as a paired slope chart, and as a scatter in which all eight models sit above the diagonal. The pairing matters—the two readers see the same forward pass on the same rows, so the right question is whether the difference is positive, and it is, on every model.

3.3 Why is the output reader not simply under-trained?

Two objections could explain the head advantage without any claim about *where* relevance lives, and we tested both.

Objection 1: the head probe gets a better read-out. The head probe selects 20 units on validation and concatenates them; the residual probe reads one 3072–4096-dimensional vector. The natural control is to give the final layer the *identical* procedure: same 2560-dimensional budget, validation-selected 20 chunks, per-unit standardisation, concatenation. Under that matched proce-

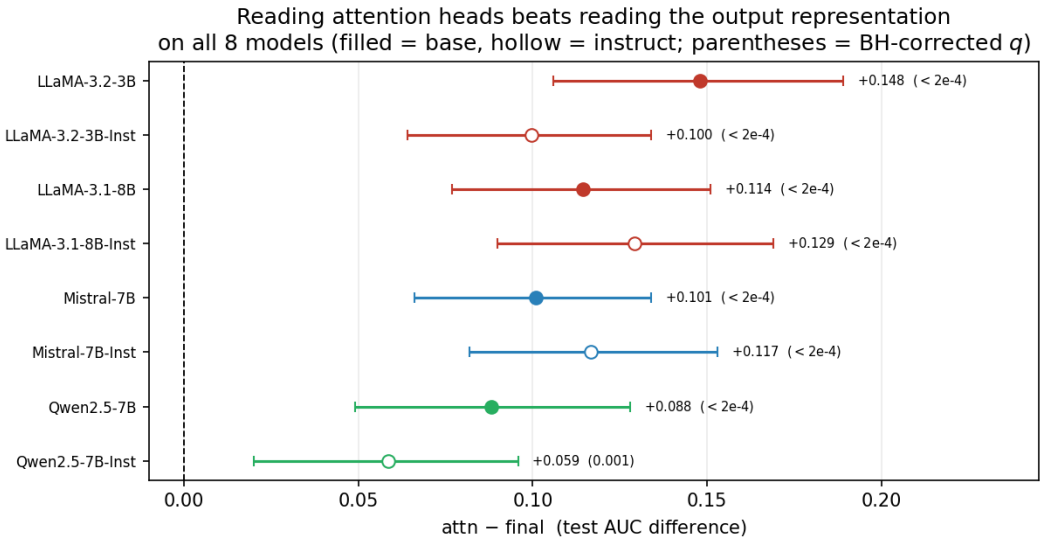


Figure 2: The attention-head edge per model. Filled markers are base models, hollow are instruct. Every interval clears zero after correction; the smallest effect (Qwen2.5-7B-Instruct, +0.059) is also the one with the largest correction penalty.

dure the edge barely moves (+0.043 to +0.128, significant on 8/8). The head advantage is therefore not bought by the read-out procedure or by the selection step—but this control cuts both ways, and we report the other direction honestly in §5.

Objection 2: the head probe is trained and the judge is not. This is correct as stated, and it is why we also compare against a *trained* reader on the output representation. That reader is the `final` column of Table 2: trained on 5218 labelled pairs, it still loses to the heads on every model. The "final − judge" row of Table 3 quantifies the training contribution on its own: after correction it is not distinguishable from zero anywhere, and its point estimates are negative on seven of eight models. Training on the output representation, in other words, is not what produces the advantage—the location is.

3.4 What the depth profile looks like

Figure 4 shows the per-layer picture for LLaMA-3.2-3B, and the same figure for all eight models is in the artifact bundle (`fig3_depth_<model>.png`). Three observations.

Relevance decodability from the residual stream rises into the middle of the network and is no longer rising at the output layer. The peak is a validation-selected layer, and it is well before the end (L9 of 28 for LLaMA-3.2-3B; the peak layers for the other models are in Table 2).

The per-head map is sparse and mid-depth. The best single head reaches 0.64–0.65 on LLaMA-3.2-3B and Mistral-7B, while the *mean* head in the same layer sits at 0.50–0.62: most heads carry little and a few carry a lot, which is why a probe that selects units on validation does well here, and why we report the selection procedure explicitly. Early layers are markedly weaker: the first layer is exactly uninformative for every single head on LLaMA-3.2-3B, and the first four are on Mistral-7B, whereas Qwen2.5-7B has informative heads from the start.

The selected heads cluster at a characteristic relative depth per family (Figure 5): mid-network for LLaMA (≈0.35–0.52 of depth), later for Qwen (whose best single head reaches 0.72 at layer 18

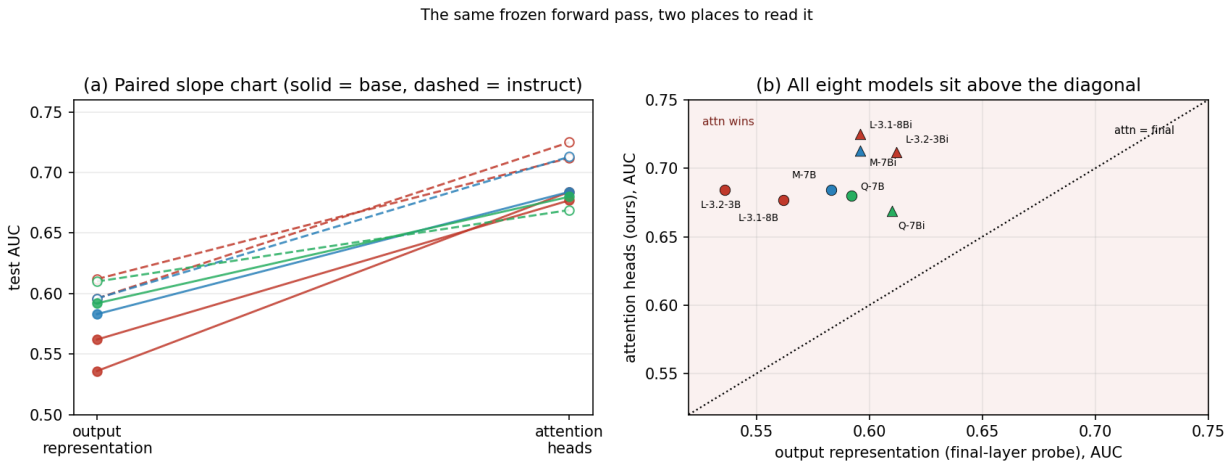


Figure 3: (a) Paired slope from the output representation to the attention heads; every line rises. (b) The same eight points as a scatter; all sit in the shaded region where the head probe wins. Colours are model families.

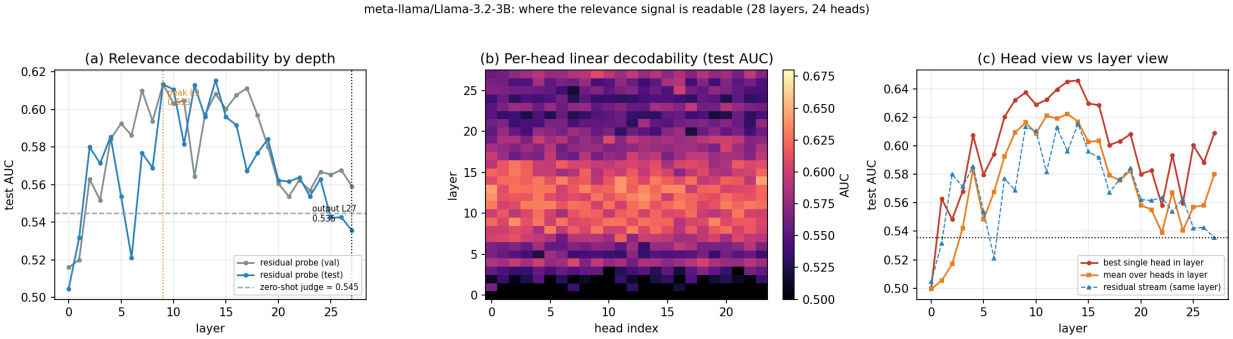


Figure 4: LLaMA-3.2-3B. (a) Residual-stream decodability by depth, against the zero-shot judge baseline; the peak is validation-selected, the output layer is marked. (b) Per-head decodability; the informative heads form a sparse, mid-depth band. (c) Best and mean head per layer versus the residual stream at the same layer. Per-model versions are in the artifact bundle.

of 28), and Mistral's informative heads appear after its uninformative early band even though its residual stream carries the signal from much earlier. We report this as description; we did not find a principled account of why the depths differ.

4 Controls and robustness

Read-out matched to the head probe. Described in §3; the edge persists at +0.043 to +0.128 on 8/8 models when the final layer gets the head probe's exact procedure.

Read-out replaced by an ensemble. We also replaced the single linear read-out of the final layer with a per-chunk logistic-regression ensemble. This *raises* the final layer substantially (e.g. 0.536 → 0.621 on LLaMA-3.2-3B, a gain of +0.085—comparable to the entire residual-stream edge). On two of eight models it closes or reverses the head advantage (Qwen2.5-7B: 0.683 vs.

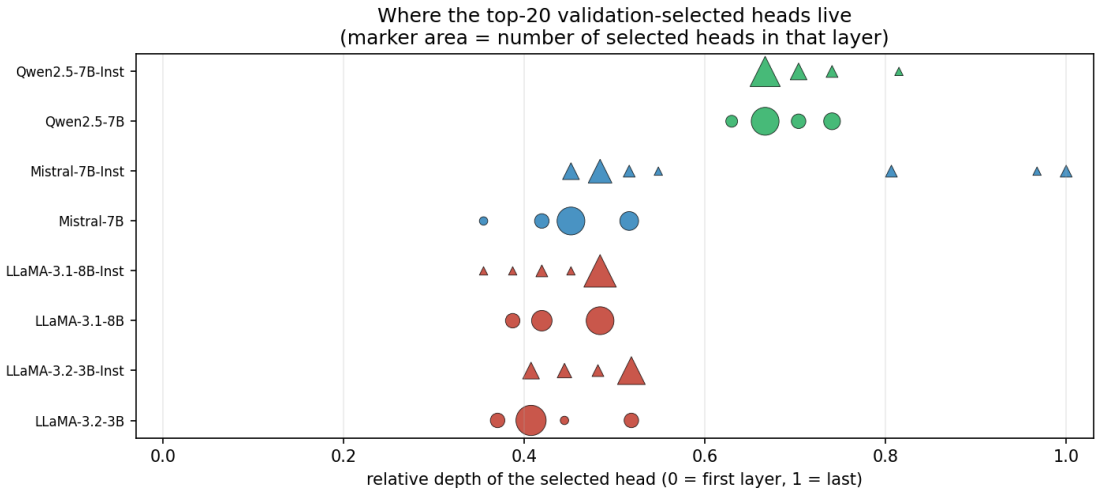


Figure 5: Where the validation-selected heads live, in relative depth. Marker area is the number of selected heads in that layer.

0.680; Qwen2.5-7B-Instruct: 0.719 vs. 0.669). We therefore do not claim that *attention heads beat the output end*; we claim the specific matched comparison of Table 2, and we flag the read-out sensitivity as a boundary in §5.

MLP read-outs. At matched parameter count and at a wider setting, final-layer MLPs reach 0.594 and 0.560 on LLaMA-3.2-3B—above the linear final-layer probe (0.536) but still below the head probe (0.684).

Prompt wording. The comparison holds under two rewritings of the prompt (a terse "Passage/Query/Relevant?" form and a verbose form). We report this as a direction check rather than as a strength claim: the verbose form is ≈17% longer in tokens, and we did not attempt to separate wording from length.

A second dataset. On FiQA with BM25 hard negatives (LLaMA-3.2-3B, 1500 queries), the head probe reaches 0.884 against a 0.724 output representation and a 0.597 judge, so the ordering is unchanged on a second in-domain dataset. This is one model, not eight, and we do not lean on it.

5 Where the claim stops

This is one specific comparison. The headline contrast is defined against a single linear read-out of the final-layer residual. We showed in §4 that a different read-out of that same layer changes the answer on two of eight models. A reader who prefers a per-chunk ensemble as "the" output representation should read our result as *read-out-dependent*, not as a general claim about the output end.

The residual-stream form of the claim is narrower. "Internal beats output representation" is often stated for the residual stream. In our panel that contrast is significant on the three base

models only, and its instruct counterparts show smaller, non-significant effects. We report the head comparison as the headline precisely because it is the one that replicates everywhere.

The judge is a zero-shot judge. We compare against the model's own verdict with no exemplars. A stronger prompt or a few-shot judge would raise the lower bound. We tested one such variant in an earlier phase and it did not close the gap, but we do not have that artifact in the current bundle, so we state the comparison as "zero-shot verdict" and nothing more.

Correlational, not causal. Every reader here is a probe. The companion work on this project attempted a causal account of the output's weakness and did not support it; the present paper therefore contains no mechanistic assertion, and readers should not infer one.

Relevance means MS MARCO relevance. Labels come from dataset qrels for one retrieval collection (plus one FiQA replication). We make no cross-domain and no reranking-accuracy claim.

6 Related work

[TODO: restore citations: this section is deliberately citation-free in this draft. The bibliography in `paper/refs.bib` already contains the entries this project verified as real, covering (i) probing classifiers, (ii) the logit/tuned lens, (iii) activation steering and representation engineering, (iv) calibration and hallucination work showing models encode correctness they do not surface, and (v) concurrent work on hallucination detection from mid-layer hidden states. The net-novelty statement to write is: internals-outstrip-the-output is not new as a slogan, but a controlled, corrected, eight-model comparison localised to *retrieval relevance*, with the head-versus-output-representation contrast as the surviving effect, is.]

7 Conclusion

Across eight models from three families, reading the top attention heads of a frozen forward pass recovers retrieval relevance better than reading the final-layer hidden state, better than any single residual layer, and far better than the model's own verdict. The margin over the output representation is +0.06 to +0.15 AUC, it holds on every model, and it is the only one of four contrasts we tested that survives multiple-comparison correction everywhere. Training is not the explanation—a probe on the output representation never beats simply asking the model. If you need to know whether a passage is relevant, the model's heads know, and its mouth is the last place to look.

A Artifact map

Every number in this paper traces to a JSON artifact, and every figure is regenerated by a script from those artifacts.
Scripts: `scripts/collect_paper_numbers.py` (paper numbers), `scripts/aggregate_significance.py` (BH correction), `scripts/plot_paper_figures.py` (all figures), `scripts/selfcheck.py` (structural self-check, including that all four prompt builders agree byte-for-byte).

Artifact	Contents
`results/PAPER_NUMBERS.json`	single source of truth, with provenance per value
`results/significance_<model>_q1500.json`	four reader AUCs + CIs + the four edges
`results/significance_summary_q1500.json`	BH-corrected edge table across models
`results/capacity_control_<model>_q1500.json`	read-out matched / ensemble / MLP controls
`results/figures/fig3_depth_<model>.png`	per-model depth + per-head figure
`results/figures/fig*.json`	provenance sidecar for every figure